

Literature-implied answer to Nygaard’s semi-embedding question

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Status. Literature-implied answer. This is not a new proof packet. The implication is agent-identified: the supporting source records the needed Tauberian/semi-embedding theorem but does not explicitly state that it is answering Nygaard’s Question 6.2.

Source question. Olav Nygaard, *Boundedness and surjectivity in normed spaces*. See arXiv:math/0009034, Question 6.2, page 13. Nygaard asks:

Motivated by Theorem 3.2 we ask the following question:

Question 6.2. *Suppose $A \subset B_X$ is thin. Is there a Banach space Y and a semi-embedding $T : Y \rightarrow X$ such that $TY \supset A$ but T is not onto.*

In words: if $A \subset B_X$ is thin, is there a Banach space Y and a semi-embedding $T : Y \rightarrow X$ such that $TY \supset A$ but T is not onto?

Supporting theorem. Jose Rodriguez, *On integration in Banach spaces and total sets*. Remark 4.7 on page 11 of arXiv:1806.10049 records the standard Neidinger–Rosenthal characterization:

Remark 4.7. Let $T : X \rightarrow Y$ be a tauberian operator (i.e. $(T^{**})^{-1}(Y) = X$). In this case, the argument of [17, Proposition 8] shows that a scalarly measurable and scalarly bounded function $f : \Omega \rightarrow X$ is Pettis integrable whenever $T \circ f$ is Pettis integrable. We stress that an operator $T : X \rightarrow Y$ is injective and tauberian if and only if the restriction $T|_Z$ is a semi-embedding for every subspace $Z \subseteq X$, see [30].

Namely, an operator $T : X \rightarrow Y$ is injective and Tauberian if and only if the restriction $T|_Z$ is a semi-embedding for every subspace $Z \subset X$. The remark cites Neidinger and Rosenthal, “Norm-attainment of linear functionals on subspaces and characterizations of Tauberian operators”, Pacific J. Math. 118 (1985), no. 1, 215–228.

Identification. Nygaard’s Theorem 3.2 proves that if $A \subset X$ is thin, then there is a Banach space Y and a Tauberian injection $J : Y \rightarrow X$ such that $JY \supset A$ but J is not onto X . Applying the Neidinger–Rosenthal characterization to J , with $Z = Y$, shows that J itself is a semi-embedding. Therefore the operator constructed in Nygaard’s proof already has the extra property asked for in Question 6.2.

Answer. Yes. Every thin $A \subset B_X$ admits a Banach space Y and a semi-embedding $T : Y \rightarrow X$ such that $TY \supset A$ and T is not onto.

Scope limitations. This packet answers only Question 6.2. It does not address Nygaard's Question 6.1 on bounded fundamental or norming thin sets, Question 6.3 on thickness of inner functions/Blaschke products, or Question 6.4 on James boundaries in H^∞ .

References

- [1] Olav Nygaard. Boundedness and surjectivity in normed spaces. arXiv:math/0009034, 2000.
<https://arxiv.org/abs/math/0009034>.
- [2] Jose Rodriguez. On integration in Banach spaces and total sets. arXiv:1806.10049, 2018.
<https://arxiv.org/abs/1806.10049>.
- [3] R. Neidinger and H. P. Rosenthal. Norm-attainment of linear functionals on subspaces and characterizations of Tauberian operators. Pacific J. Math. 118 (1985), no. 1, 215–228.